

Smart Workout: A RAG-Assisted Decision Support and Business Intelligence System for Personalized Weight Training

Final report

by

Aye Khin Khin Hpone (st125970)

Waranon Neamtuptim (st125934)

Jirapat Datephanyawat (st125843)

**AT82.04: Business Intelligence and Analytics
Intersem 2026**

Dr. Vatcharaporn Esichaikul

TA: Anushka Ojha

Asian Institute of Technology
Faculty of Advanced Science and Technology (FAST)
Thailand
5 July 2026

ABSTRACT

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

Weight training is widely practiced for strength and fitness, yet many gym users struggle to plan effectively: they are uncertain which exercises suit a given body part, how to adapt to available equipment, how to structure a weekly timetable, and whether their physical condition supports a particular plan. Most decisions rely on generic online routines that ignore personal factors — age, sleep, BMI, physiological indicators, equipment, schedule — producing a mismatch between plan and readiness. This project proposes Smart Workout, a combined Decision Support System (DSS) and Business Intelligence System (BIS) (Power, 2002; Chaudhuri et al., 2011) that unifies multi-source real-world data, a Retrieval-Augmented Generation (RAG) knowledge layer, and supervised ML models within an interactive web dashboard, transforming user-provided inputs into descriptive insight, predictive outcome estimates, and prescriptive training plans while preserving human oversight. The primary target users are gym members seeking personalized training guidance; fitness coaches constitute a secondary audience using the dashboard for decision-support reference.

1.2 Objectives of the Study

The project is expected to deliver a functional DSS/BIS prototype that fulfils the following objectives, explicitly covering the descriptive, predictive, and prescriptive analytics layers required of a BI system:

- **Data collection & preparation:** clean and standardize multi-source public datasets (gym membership and check-in records, exercise catalogue, gym-member workout tracking, sleep & lifestyle, daily nutrition; details in §4) alongside structured user input.
- **Descriptive analytics:** segment users into lifestyle profiles via clustering; present a relational gym membership view (member activity, subscription trends, usage analytics) as an interactive dashboard; and present cross-dataset insights (calorie-burn patterns, exercise coverage, nutrition macro mix).
- **Predictive analytics:** train and compare multiple candidate models per task via cross-validation: (a) calorie-burn-rate regression (kcal/hr); (b) Workout-intensity 3-class clas-

sification (Low/Mid/High).

- **Prescriptive analytics:** generate a daily/weekly training plan via a rule-based engine conditioned on the predicted intensity, readiness, target body part, and available equipment; the predicted calorie-burn rate is displayed alongside the plan. A rule-derived Training Readiness band (computed from user input; rules drawn from the Sleep Health dataset) guides the plan generator (see §5).
- **RAG knowledge retrieval & AI chat assistant:** retrieve exercise instructions, equipment-substitution rules, and scheduling guidance via an interactive chat interface (RAG retrieval is core; LLM chat is a planned extension).
- **BI dashboard:** React-based dashboard (with embedded Tableau views) presenting a relational gym membership view, multi-dataset descriptive analytics, personalized predictions, prescribed plan, RAG-sourced explanation, and embedded chat.

CHAPTER 2

LITERATURE REVIEW

The concept of leveraging data-driven systems to enhance physical training has gained significant traction in recent years. In this chapter, an outline of the foundational theories regarding Decision Support Systems (DSS), Business Intelligence (BI), and modern Machine Learning (ML) techniques will be presented. This review is divided into four primary sections: the architecture of BI systems in fitness, the application of predictive modeling for physiological outcomes, the integration of Retrieval-Augmented Generation (RAG) for knowledge grounding, and the impact of lifestyle factors on training readiness. These discussions provide the necessary theoretical framework to support the development of the Smart Workout system.

2.1 Decision Support and Business Intelligence Systems

Decision Support Systems (DSS) are defined as interactive computer-based systems intended to help decision-makers use communications technologies, data, documents, knowledge, and models to identify and solve problems. According to Power [Power, 2002], these systems are essential for managers and individuals to process complex information that exceeds human cognitive limits.

In the context of the fitness industry, Business Intelligence (BI) technology enables the integration of multi-source data to provide actionable insights. Chaudhuri et al. [Chaudhuri et al., 2011] emphasize that effective BI systems should support descriptive, predictive, and prescriptive analytics. The proposed Smart Workout system adopts this three-layer approach by transforming raw gym membership and check-in records into descriptive insights, predictive analytics, and prescriptive training recommendations.

2.2 Machine Learning in Exercise Analytics

The predictive layer of the Smart Workout system relies on supervised machine learning models to estimate physiological responses such as calorie-burn rates and workout intensity. Modern research frequently utilizes gradient-boosted trees, such as XGBoost, for high-performance classification and regression tasks. Chen and Guestrin [Chen and Guestrin, 2016] demonstrated that XGBoost provides a scalable and accurate tree-boosting framework that outperforms traditional linear baselines on complex datasets.

Furthermore, interpretability is a critical requirement for health-related decision support systems to improve user trust and transparency. The SHAP (SHapley Additive exPlanations) method, proposed by Lundberg and Lee [Lundberg and Lee, 2017], enables feature-attribution analysis by quantifying the contribution of each input feature to a model’s prediction. In the proposed Smart Workout system, SHAP is used to explain how factors such as demographics and heart rate influence calorie-burn estimation and workout-intensity recommendations.

2.3 Retrieval-Augmented Generation (RAG)

While large language models (LLMs) possess vast parametric memory, they often struggle with factual accuracy and hallucinations when providing specific instructions. To address this, Lewis et al. [Lewis et al., 2020] introduced Retrieval-Augmented Generation (RAG), a technique that retrieves relevant document snippets from a vector store to ground the model’s responses. In the Smart Workout system, this technology is used to index exercise catalogues (ExerciseDB) and curated training rules to provide grounded rationale and equipment substitutions. By utilizing sentence embeddings to match user queries with verified fitness data, the system ensures that the prescriptive plans are supported by evidence rather than generated in a vacuum.

2.4 Lifestyle Factors and Training Readiness

Effective physical training cannot be decoupled from an individual’s recovery and lifestyle patterns. Research into sleep health and stress levels has shown that these factors significantly influence an individual’s readiness to perform high-intensity exercise. The use of K-Means clustering allows for the identification of lifestyle archetypes, which categorize users based on their sleep duration, activity levels, and stress scores. By integrating these descriptive profiles into the prescriptive engine, the system can modulate training volume and load to match the user’s current physical condition. This holistic approach ensures that the generated workout plans are not only personalized to users’ goals but also adaptive to daily physiological constraints.

2.5 Chapter Summary

Presented in this chapter are the core concepts and technological approaches related to the problem of generic and non-adaptive workout planning. There are several methods to tackle this problem, including traditional coaching and static online routines, but these often fail to account for real-time personal data. Among the discussed methods, the combination of Business Intelligence (BI) layers and Retrieval-Augmented Generation (RAG)-grounded AI has the advantage of providing both high-level analytics and specific,

evidence-based guidance. It has been suggested by various researchers that integrating structured data with generative models can significantly improve the reliability of decision support [Power, 2002, Lewis et al., 2020]. The following chapter outlines the research methodology and system architecture implemented to realize this integrated framework.

CHAPTER 3

METHODOLOGY

This chapter outlines the systematic approach used to develop the Smart Workout Decision Support System (DSS) and Business Intelligence System (BIS). It describes the multi-source data integration process, the machine learning pipeline for predictive modeling, and the architecture of the Retrieval-Augmented Generation (RAG) layer. Furthermore, it explains the development of the rule-based engine and the interactive dashboard interface, demonstrating how raw data is transformed into prescriptive training plans. The following sections present the data-to-decision workflow that forms the foundation of this research.

3.1 Research Framework and System Architecture

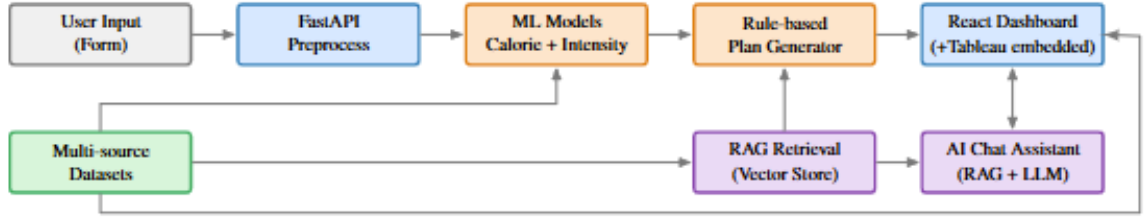


Figure 1: Overall architecture of the proposed Smart Workout Decision Support System.

The Smart Workout system is designed around a three-layer Business Intelligence (BI) framework consisting of descriptive, predictive, and prescriptive analytics. As shown in Figure 1, the process begins with a user input form which is processed via a FastAPI back-end. The architecture integrates five distinct datasets to support these layers, ensuring that the system moves beyond mere data visualization to providing actionable guidance. This modular design allows for the independent validation of each stage, from initial data pre-processing to final plan generation and RAG-grounded explanations.

3.2 Data Sources and Preprocessing

The research utilizes five real-world public datasets to ground the system in diverse fitness and lifestyle contexts, as summarized in Table 1. These include the Gym Database Management System for membership analytics (305,013 records), ExerciseDB for cataloging 1,500 exercises, and the Gym Members Exercise Tracking dataset (973 records) for training predictive models. Additionally, the Sleep Health & Lifestyle and Daily Food Nutrition datasets provide the necessary context for user profiling and nutrition-aware guidance. All raw data underwent rigorous inspection, cleaning, and standardization to ensure consistency

across the relational star schema and the vector store used for retrieval.

Table 1: Summary of data sources used in the Smart Workout system.

Source table/file	Records	Role in the system
Gym DB – Users (Amoghabhinavms, n.d.)	5,000	Member dimension table containing demographics, signup dates, locations, and subscription-plan keys. Supports dashboard filters and member segmentation.
Gym DB – Check-in/check-out history (Amoghabhinavms, n.d.)	300,000	Central session fact table containing gym visits, workout type, timestamps, duration-derived features, and calories burned. Primary descriptive and warehouse source.
Gym DB – Subscription plans (Amoghabhinavms, n.d.)	3	Plan dimension table containing monthly price and feature descriptions. Supports subscription mix and estimated recurring-revenue summaries.
Gym DB – Gym locations (Amoghabhinavms, n.d.)	10	Location dimension table containing branch city, gym type, and facilities. Supports utilization analysis and location-level plan constraints.
ExerciseDB (raw flat) (ExerciseDB, n.d.)	1,500	Exercise catalogue (body parts, equipment, target muscles, instructions). Supports the exercise recommender and RAG pipeline.
Gym Members Exercise Tracking (Khorasani, 2024)	973	Workout sessions with calories, BPM, duration, workout type, BMI, experience. Primary ML training data.
Sleep Health & Lifestyle (Tharmalingam, 2023)	374	Informs lifestyle clustering and <i>Training Readiness</i> rules derived from user input.
Daily Food Nutrition (Shamim, 2025)	651	Macro/calorie context on the dashboard; nutrition-aware snippets embedded in RAG.

3.3 Data Architecture and Schema Design

The integration of multi-source datasets within the Smart Workout system requires a structured architectural design to ensure data consistency across the descriptive, predictive, and prescriptive layers. This section details the logical and physical relationships between the various data components, ranging from the core relational gym database to the isolated datasets used for lifestyle clustering and RAG retrieval. The following diagrams illustrate the complexity of the data ecosystem and how independent sources are unified through either hard foreign keys or logical joins to provide a comprehensive user profile.

As shown in Figure 2, the system utilizes an eight-dataset schema that categorizes information into connected and isolated groups. The connected group forms the primary gym star schema, which tracks user demographics, subscription plans, and check-in history to support membership analytics. Conversely, isolated datasets such as ExerciseDB, Sleep Health & Lifestyle, and Daily Food Nutrition are integrated via logical joins based on attributes such as workout type, age, and BMI. This design allows the system to combine macro-level

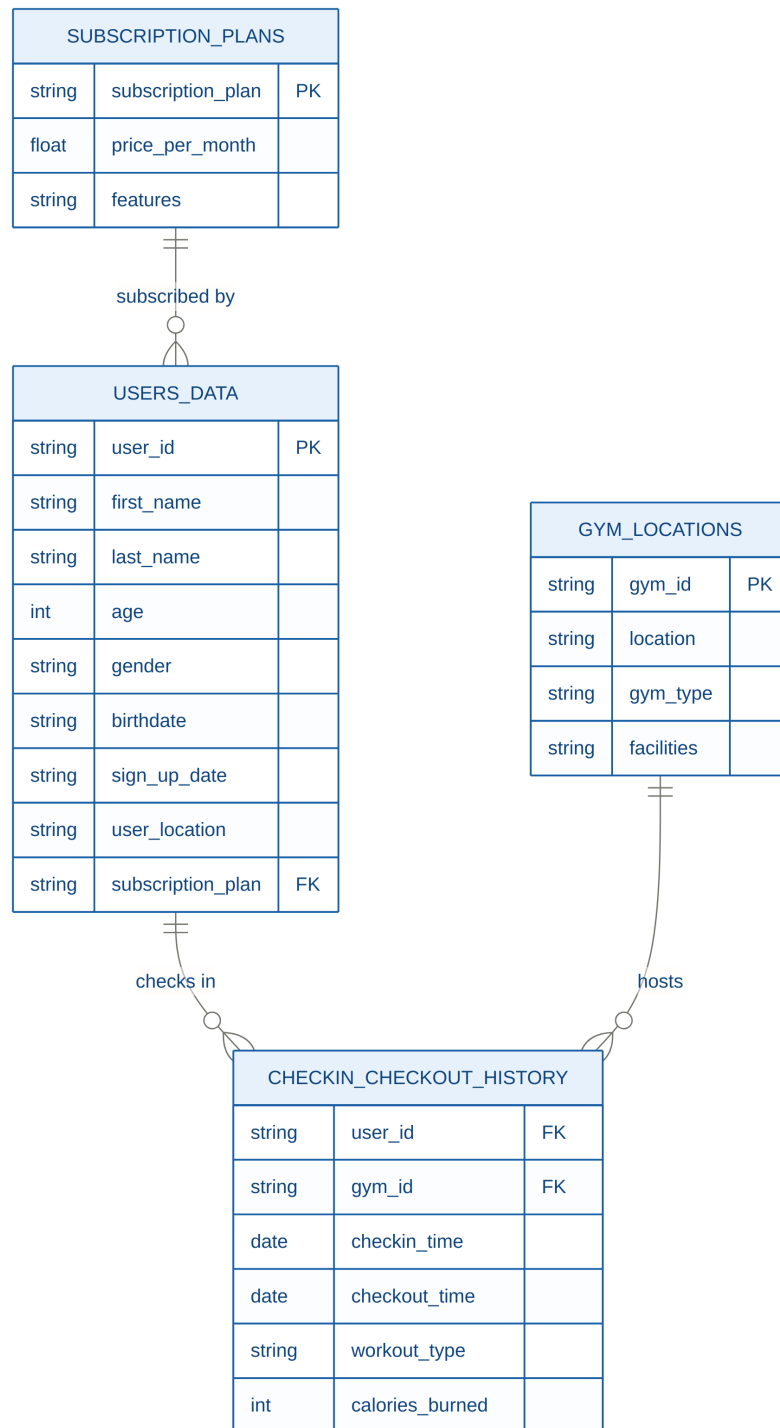


Figure 2: Integrated data architecture of the Smart Workout system showing connected and isolated datasets.

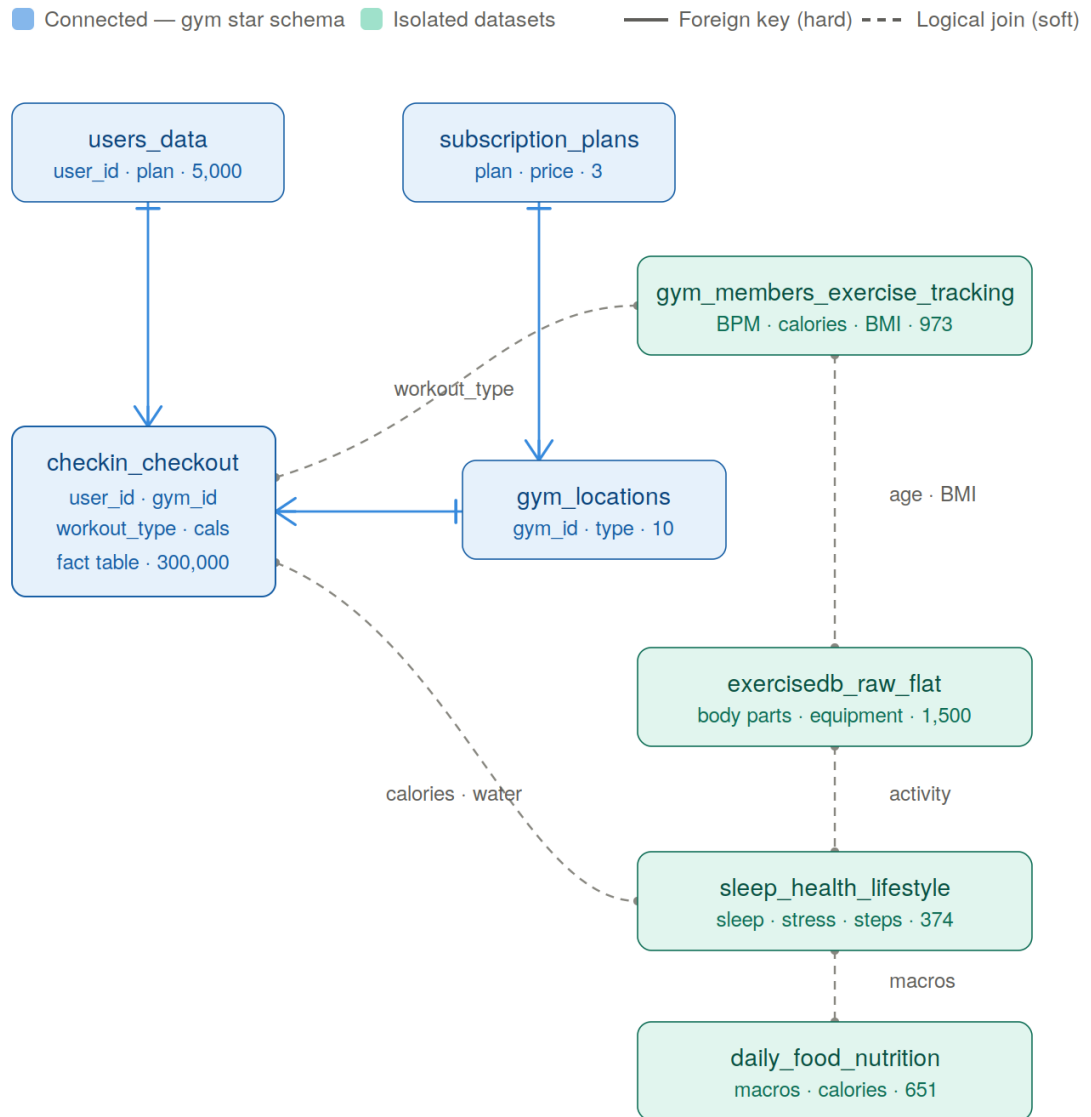


Figure 3: Relational star schema of the Gym Database Management System.

lifestyle insights with micro-level exercise instructions to support evidence-based training plan generation.

The core of the descriptive analytics layer is supported by a relational star schema specifically designed for the Gym Database Management System, as illustrated in Figure 3. This schema centers on the `CHECKIN_CHECKOUT_HISTORY` fact table, which links to three primary dimension tables: `USERS_DATA`, `SUBSCRIPTION_PLANS`, and `GYM_LOCATIONS`. By utilizing primary and foreign key relationships (e.g., `user_id`, `gym_id`, and `subscription_plan`), the system can efficiently query more than 300,000 session records to identify utilization trends and subscription mix patterns. This relational foundation ensures that the BI dashboard provides a high-fidelity view of gym operations and member behavior.

3.4 Predictive Modeling Pipeline

The predictive layer employs a best-of-three model selection strategy for two primary tasks: calorie-burn-rate regression and workout-intensity classification. For the regression task, candidates include Ridge Regression, Random Forest Regressor, and XGBoost Regressor, with selection based on minimizing Root Mean Square Error (RMSE). The classification task evaluates Multinomial Logistic Regression, Random Forest, and XGBoost to categorize intensity into three levels (Low, Mid, High). Models are evaluated using 5-fold cross-validation to ensure generalizability, and SHAP (SHapley Additive exPlanations) is applied to the selected models to provide transparency regarding which physiological features most heavily influence the predictions.

3.5 RAG and Prescriptive Plan Generation

The prescriptive layer combines a rule-based engine with a Retrieval-Augmented Generation (RAG) knowledge layer to generate personalized training plans. The RAG system utilizes ChromaDB and sentence embeddings (all-MiniLM-L6-v2) to index exercise instructions and equipment substitution rules. When a user submits their profile, the rule-based generator integrates the predicted intensity, the calculated training readiness band, and target body parts to filter the ExerciseDB pool. This ensures that the resulting daily or weekly plans are not only specific to the user's goals but are also grounded in verified evidence retrieved from the knowledge base.

3.6 System Implementation and Deliverables

The final implementation is delivered as a full-stack web application utilizing React for the frontend and FastAPI for the backend orchestration. The dashboard features embedded Tableau views for relational membership analytics and lifestyle profiling, alongside cus-

tom React components for displaying prediction results and the generated workout plan. Key deliverables of this methodology include the working prototype, the validated machine learning endpoints, and the RAG-assisted chat interface. This integrated technology stack enables the system to provide a seamless transition from descriptive insight to predictive estimation and, ultimately, prescriptive action.

3.7 Chapter Summary

Presented in this chapter is the comprehensive methodology for constructing a data-driven decision support tool for weight training. The methods described, ranging from multi-source data cleaning to the deployment of boosted-tree models and vector databases, address the limitations of generic, non-adaptive training plans identified in earlier chapters. Among these methods, the integration of RAG-grounded AI provides the distinct advantage of delivering evidence-based explanations alongside prescriptive plans. These methodological steps ensure that the results and discussions presented in Chapter 4 are based on a robust, validated, and transparent system architecture.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Final Synthesised Ranking

CHAPTER 5

CONCLUSION AND RECOMMENDATIONS

5.1 Conclusion

5.2 Recommendations

5.3 Future Research Directions

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